SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: MLA0303
COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr DUMPALA PRASANTH

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario-Based Assessment **DURATION: 1.5 Hrs**

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario.
(5 Marks)

2. A warehouse robot uses Reinforcement Learning to find the shortest path for delivering packages while avoiding obstacles. Sometimes the robot chooses longer paths due to insufficient exploration during training. Solve how the exploration vs exploitation trade-off affects the robot's navigation performance and suggest some improvements. **(5 Marks)**

3. A hospital is considering using Reinforcement Learning to optimize patient appointment scheduling in order to reduce waiting time and improve resource utilization. Demonstrate the suitability of RL for this scenario and discuss possible challenges such as delayed rewards and ethical considerations. **(5 Marks)**

4. A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather

conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm.

(5 Marks)

5. An online gaming platform uses Reinforcement Learning to match players of similar skill levels to maintain engagement. Report how reward design can influence matchmaking quality with reward definition that might lead to biased or inefficient matching results. **(5 Marks)**

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

(5 Marks)

7. A drone delivery company uses RL to optimize delivery routes while avoiding restricted airspace. Construct a suitable RL framework by defining states, actions, and rewards, and justify how penalties can ensure safe navigation. **(5 Marks)**

8. A university applies RL to personalize online learning paths for students. Examine how reward design can influence student engagement and defend the ethical considerations of adapting learning content dynamically. **(5 Marks)**

9. A traffic management authority deploys RL to control traffic lights at busy intersections. Illustrate how constraints such as emergency vehicle priority and pedestrian safety can be incorporated into the RL model and interpret the impact on congestion reduction. **(5 Marks)**

10. A manufacturing plant uses RL to schedule machines under resource limitations. Differentiate between traditional optimization methods and constrained RL approaches, and assess which method provides better adaptability in dynamic production environments. **(5 Marks)**

Code of Conduct:

I __**K Uday Kiran Reddy (192425211)**__ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: kuday

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1. Personalized Notification System Using Reinforcement Learning

Scenario

A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns.

Answer

Reinforcement Learning can be used to learn the best notification strategy by observing user behavior and receiving rewards based on how users respond to notifications.

RL Framework

RL Component	Definition
Agent	Mobile application notification system
Environment	User and mobile app environment
State	User activity, time of day, previous notification response, location, app usage frequency
Actions	Send notification, delay notification, change notification type, or do not send
Reward	Positive reward for opening/clicking notification and negative reward for ignoring or disabling notifications

Possible States

- User is active or inactive.
- Time of day.
- Frequency of recent app usage.
- Previous notification response.
- Type of notification preferred by the user.
- Number of notifications received recently.

Possible Actions

1. Send notification immediately.
2. Delay notification.
3. Send a personalized notification.
4. Do not send notification.

Reward Function

A suitable reward function can be:

$$R = \begin{cases} +10 & \text{if user opens notification} \\ +5 & \text{if user clicks the notification} \\ -3 & \text{if notification is ignored} \\ -10 & \text{if user disables notifications} \end{cases}$$

The agent learns a policy that maximizes the long-term reward.

Suitable Algorithm

A **policy-based method such as REINFORCE** can learn the probability of selecting different notification actions.

Conclusion

RL can continuously adapt notification timing and content according to individual user behavior. This can increase engagement while reducing unnecessary notifications.

Question 2. Warehouse Robot Navigation and Exploration–Exploitation

Scenario

A warehouse robot uses RL to find the shortest path for package delivery while avoiding obstacles. Sometimes it chooses longer paths due to insufficient exploration.

Answer

The exploration–exploitation trade-off determines whether the robot tries new routes or uses routes that it already knows.

Exploration

Exploration means selecting unfamiliar paths to discover potentially shorter or safer routes.

Exploitation

Exploitation means selecting the best-known route based on previous experience.

If the robot performs **too little exploration**, it may repeatedly use a longer route and never discover a better path.

Example

Suppose the robot has learned:

- Route A = 15 seconds
- Route B = 10 seconds
- Route C = unknown

If the robot always exploits Route B, it may never explore Route C, even though Route C might take only 7 seconds.

Improvements

1. **Epsilon-greedy strategy**
   Select the best action most of the time but occasionally explore another action.
2. **Decaying epsilon**
   Use high exploration during early training and gradually reduce exploration.
3. **Reward shaping**
   Give positive rewards for reaching the destination quickly and penalties for collisions and unnecessary movements.
4. **Safe exploration**
   Prevent the robot from exploring actions that can cause collisions.
5. **Experience replay**
   Store previous experiences and reuse them during training.

Reward Example

$$R = +100 \text{ for successful delivery}$$

$$R = -50 \text{ for collision}$$

$$R = -1 \text{ for every unnecessary movement}$$

Conclusion

A balanced exploration–exploitation strategy allows the robot to discover efficient routes during training while eventually exploiting the safest and shortest route.

Question 3. RL for Hospital Appointment Scheduling

Scenario

A hospital wants to use RL to optimize patient appointment scheduling, reduce waiting time, and improve resource utilization.

Answer

RL is suitable because appointment scheduling involves sequential decisions where the current decision can affect future waiting times, doctor availability, and resource utilization.

RL Model

Component	Description
Agent	Hospital scheduling system
Environment	Hospital, patients, doctors and resources
State	Waiting patients, doctor availability, appointment queue, emergency cases
Action	Assign patient to doctor/time slot
Reward	Reduced waiting time and improved resource utilization

Reward Function

$$R = -(\text{Patient Waiting Time}) + (\text{Resource Utilization})$$

Additional penalties can be provided for:

- Appointment conflicts.
- Excessive patient waiting.
- Unused resources.
- Incorrect prioritization of emergency cases.

Delayed Rewards

Some decisions may not show their effect immediately. For example, scheduling a patient earlier may cause a queue later in the day. Therefore, the system must consider **long-term cumulative rewards** rather than only immediate rewards.

Challenges

1. Patient information privacy.
2. Ethical treatment of patients.
3. Emergency cases must receive priority.
4. Incorrect policies could negatively affect patient care.
5. Training data may contain historical scheduling bias.

Conclusion

RL can improve hospital scheduling efficiency, but it should operate with safety constraints and human supervision. Patient safety must always take priority over optimization.

Question 4. Smart Home Energy Management Using RL

Scenario

A smart home automatically controls lights, fans, and air conditioners based on user presence and weather conditions.

Answer

RL can learn an energy-efficient control policy by observing household conditions and receiving rewards based on energy consumption and user comfort.

RL Model

Agent: Smart home controller

Environment: Home appliances and surrounding conditions

State Space

The state can contain:

- Number of people present.
- Indoor temperature.
- Outdoor temperature.
- Humidity.
- Time of day.
- Current electricity consumption.
- Appliance status.

Action Space

The controller can:

- Turn lights ON/OFF.
- Turn fan ON/OFF.
- Increase/decrease AC temperature.
- Turn appliances to energy-saving mode.

Reward Structure

$$R = -\text{Energy Consumption} - \text{Discomfort Penalty}$$

For example:

- Lower electricity consumption → positive reward.
- Comfortable temperature → positive reward.
- Excessive electricity use → negative reward.
- Turning off necessary appliances → comfort penalty.

Suitable Algorithm

A policy-based algorithm such as **PPO (Proximal Policy Optimization)** can be used because it provides stable policy updates and can handle multiple possible actions.

Conclusion

The RL controller can learn user preferences and automatically balance **energy savings and comfort**, reducing electricity consumption without significantly affecting the user's lifestyle.

Question 5. RL-Based Online Game Matchmaking

Scenario

An online gaming platform uses RL to match players of similar skill levels to maintain engagement.

Answer

The RL agent can learn matchmaking decisions by observing player skill, previous matches, win/loss history, waiting time, and player satisfaction.

RL Components

Component	**Example**
State	Player skill, win rate, waiting time, experience
Action	Match players or create a different pairing
Reward	Fair matches, player satisfaction and continued engagement

Reward Design

A suitable reward could be:

$$R = \alpha(\text{Fairness}) + \beta(\text{Engagement}) - \gamma(\text{Waiting Time})$$

A good match should have:

- Similar skill levels.
- Reasonable waiting time.
- Balanced probability of winning.
- High player satisfaction.

Problem with Poor Reward Design

Suppose the system rewards only **long playing time**.

The algorithm may intentionally create frustrating matches because frustrated players might continue playing to recover their losses.

Another problem occurs if the reward only maximizes quick matchmaking. The system may match players with very different skill levels simply to reduce waiting time.

Improvements

1. Include fairness in the reward.
2. Penalize large skill differences.
3. Consider player satisfaction.
4. Penalize excessive waiting time.
5. Monitor the system for systematic bias.

Conclusion

Reward design strongly influences matchmaking quality. A multi-objective reward that balances **fairness, engagement, skill similarity and waiting time** produces better matches than optimizing only one factor.

Question 6. RL-Based Financial Trading

Scenario

A financial trading system uses RL to decide buy/sell actions based on market signals.

Answer

RL can learn trading policies by observing market information and receiving rewards based on portfolio performance.

State

The state can include:

- Current stock price.
- Historical prices.
- Trading volume.
- Moving averages.
- Market indicators.
- Current portfolio balance.
- Current holdings.
- Market volatility.

Actions

The trading agent can:

1. Buy.
2. Sell.
3. Hold.

Reward

A basic reward can be based on change in portfolio value:

$$R_t = V_{t+1} - V_t$$

Transaction costs and risk penalties should also be considered.

$$R_t = \text{Profit} - \text{Transaction Cost} - \text{Risk Penalty}$$

Delayed Rewards

Trading decisions may not produce immediate results. A buy action could initially lose money but become profitable later.

In volatile markets:

- Rewards can change rapidly.
- Market conditions can change unexpectedly.
- Long-term consequences may be difficult to predict.
- Delayed rewards can make credit assignment difficult.

Risk Management

The system should include:

- Maximum position limits.
- Stop-loss rules.
- Transaction-cost modelling.
- Risk penalties.
- Human monitoring.

Conclusion

RL can learn complex trading strategies, but financial markets are highly uncertain. Therefore, the objective should not be profit alone; **risk-adjusted performance and safety constraints** must also be included.

Question 7. RL-Based Drone Delivery Route Optimization

Scenario

A drone delivery company wants to optimize delivery routes while avoiding restricted airspace.

Answer

RL can enable a drone to learn safe and efficient routes through repeated interaction with its environment.

RL Framework

Agent: Drone navigation controller

Environment: Airspace, weather, buildings, delivery locations and restricted zones.

State

The state may include:

- Current GPS position.
- Destination.
- Altitude.
- Battery level.
- Weather conditions.
- Nearby obstacles.
- Distance from restricted zones.
- Wind speed.

Actions

The drone can:

- Move forward.
- Move backward.
- Move left/right.
- Increase/decrease altitude.
- Change direction.
- Return to base.

Reward

$$R = \text{Delivery Reward} - \text{Distance Cost} - \text{Safety Penalties}$$

Example:

- Successful delivery: +100
- Short efficient route: +20
- Entering restricted area: -200
- Collision: -500

- Excessive battery usage: -20

Role of Penalties

Large penalties discourage the agent from entering restricted airspace or approaching dangerous obstacles.

A safety constraint can be represented as:

$$d_{\text{restricted}} > d_{\text{safe}}$$

where $d_{\text{restricted}}$ is the distance from the restricted zone and d_{safe} is the minimum safe distance.

Conclusion

RL can optimize drone routes while safety penalties and hard constraints prevent dangerous navigation. In real deployment, safety rules should override the learned policy whenever necessary.

Question 8. Personalized Online Learning Using RL

Scenario

A university uses RL to personalize online learning paths for students.

Answer

RL can recommend learning materials according to student performance, interests and learning progress.

RL Components

Agent: Online learning recommendation system

Environment: Student learning platform

State

The state can include:

- Student performance.
- Quiz scores.
- Completed topics.
- Learning speed.
- Number of attempts.
- Time spent on learning materials.
- Current difficulty level.

Actions

The agent can recommend:

1. Easy learning material.
2. Advanced material.

3. Revision.
4. Quiz.
5. Video explanation.
6. Practical exercise.

Reward

A suitable reward function could be:

$$R = \alpha(\text{Learning Progress}) + \beta(\text{Engagement}) - \gamma(\text{Dropout Risk})$$

Positive rewards can be provided when students:

- Complete lessons.
- Improve quiz scores.
- Successfully master concepts.

Influence of Reward Design

If the system rewards only **time spent on the platform**, it may recommend unnecessarily long content.

If it rewards only **quiz scores**, it may repeatedly provide easy questions rather than challenging students.

Therefore, the reward should balance:

- Learning achievement.
- Engagement.
- Appropriate difficulty.
- Long-term knowledge retention.

Ethical Considerations

1. Student data privacy must be protected.
2. The system should not discriminate between students.
3. Students should have control over personalization.
4. Important educational decisions should not depend entirely on an automated system.
5. Recommendations should be explainable.

Conclusion

RL can provide personalized learning paths, but reward design must focus primarily on **actual learning outcomes rather than simple engagement metrics**.

Question 9. RL for Traffic Signal Control

Scenario

A traffic management authority deploys RL to control traffic lights at busy intersections.

Answer

RL can optimize traffic signal timing by observing traffic conditions and selecting appropriate signal phases.

RL Components

Agent: Traffic signal controller

Environment: Road intersection

State

The state may include:

- Number of vehicles in each lane.
- Queue length.
- Waiting time.
- Current traffic-light phase.
- Pedestrian crossing status.
- Emergency vehicle presence.

Actions

The agent can:

- Keep the current signal.
- Switch to another phase.
- Extend green time.
- Reduce green time.

Reward

$$R = -(\text{Total Waiting Time} + \text{Queue Length})$$

Additional positive rewards can be given for efficient traffic flow.

Emergency Vehicle Constraint

If an ambulance or fire truck is detected, the normal RL policy can be overridden to provide priority.

Example:

$$\text{Emergency detected} \Rightarrow \text{Emergency priority mode}$$

Pedestrian Safety

The system must:

- Provide minimum pedestrian crossing time.

- Prevent conflicting vehicle and pedestrian signals.
- Maintain a safe clearance interval.
- Avoid rapidly changing signals.

These can be implemented as **hard constraints** rather than relying only on rewards.

Impact on Congestion

The RL system can reduce:

- Vehicle waiting time.
- Queue length.
- Unnecessary red-light duration.
- Fuel consumption.

Conclusion

RL can dynamically adapt traffic signals to changing traffic conditions. However, emergency priority and pedestrian safety should be treated as mandatory constraints that cannot be violated for the sake of reducing congestion.

Question 10. Traditional Optimization vs Constrained RL in Manufacturing

Scenario

A manufacturing plant uses RL to schedule machines under resource limitations.

Answer

Machine scheduling involves deciding which machine should process which job and at what time. Both traditional optimization and RL can solve this problem, but they differ in adaptability.

Traditional Optimization

Traditional methods include:

- Linear Programming.
- Integer Programming.
- Mixed Integer Programming.
- Genetic Algorithms.
- Constraint Programming.

These methods typically require a clearly defined mathematical model.

Constrained Reinforcement Learning

Constrained RL learns a policy while satisfying resource and safety constraints.

The objective can be expressed as:

$$\max_{\pi} E_{\pi}\left[\sum_{t=0}^{T} \gamma^{t} R_{t}\right]$$

subject to:

$$E_{\pi}\left[\sum_{t=0}^{T} \gamma^{t} C_{t}\right] \leq C_{\max}$$

where $C_t$represents resource or constraint costs.

Comparison

Feature	**Traditional Optimization**	**Constrained RL**
Model requirement	Usually requires explicit model	Can learn from interaction/data
Dynamic changes	May require re-optimization	Can adapt through learning
Constraints	Strongly supported	Can incorporate constraints
Training	Usually mathematical computation	Requires training experience
Adaptability	Moderate	High
New situations	May require recalculation	Can learn improved policies
Interpretability	Often easier	Can be more complex

Example

Suppose a machine suddenly becomes unavailable.

A traditional optimization system may need to solve the scheduling problem again.

An RL system can potentially adapt its learned policy and select another machine based on the new state.

Decision

For a **stable production environment**, traditional optimization can provide highly accurate schedules.

For a **dynamic production environment** where machine availability, demand and resource conditions change frequently, constrained RL can provide better adaptability.

Conclusion

Constrained RL is preferable for highly dynamic manufacturing environments because it can learn adaptive scheduling policies while considering resource limitations. However, traditional optimization remains valuable for highly structured problems where exact solutions and strict constraints are essential.